



Figure 10: Comparison of per-token decoding latency and memory usage between the sliding window approach with re-computation baseline and StreamingLLM, plotted against the cache size (attention window size) on the X-axis. StreamingLLM delivers a remarkable speedup of up to $22.2\times$ per token and retains a memory footprint similar to the re-computation baseline.

4.4 ABLATION STUDIES

Numbers of Initial Tokens. In Table 2, we ablate the effect of adding varying numbers of initial tokens with recent tokens on the streaming perplexity. The results show the insufficiency of introducing merely one or two initial tokens, whereas a threshold of four initial tokens appears enough, with subsequent additions contributing marginal effects. This result justifies our choice of introducing 4 initial tokens as attention sinks in StreamingLLM.

Cache Sizes. In Table 6, we evaluate cache size’s impact on StreamingLLM’s perplexity. Contrary to intuition, increasing the cache size doesn’t consistently lower the language modeling perplexity. This inconsistency shows a potential limitation where these models might not maximize the utility of the entire context they receive. Future research efforts should target enhancing these models’ capabilities to utilize extensive contexts better.

4.5 EFFICIENCY RESULTS

We benchmark StreamingLLM’s decoding latency and memory usage against the sliding window with re-computation, which is the only baseline with acceptable quality. Both methods are implemented using the Huggingface Transformers library (Wolf et al., 2020) and tested on a single NVIDIA A6000 GPU using the Llama-2-7B and Llama-2-13B models. As shown in Figure 10, as the cache size increases, StreamingLLM’s decoding speed has a linear growth. The sliding window with re-computation baseline has a quadratic rise in decoding latency. Thus, StreamingLLM achieves an impressive speedup, reaching up to $22.2\times$ per token. Despite its reduced latency, StreamingLLM sustains a memory footprint consistent with the re-computation baseline.

Table 6: Effects of cache size on StreamingLLM’s performance. Increasing the cache size in StreamingLLM doesn’t consistently yield a decrease in perplexity, showing these models may not fully utilize the provided context. Cache config $x+y$ denotes adding x initial tokens with y recent tokens. Perplexity is evaluated on 400K tokens in the concatenated PG19 test set.

Cache	4+252	4+508	4+1020	4+2044
Falcon-7B	13.61	12.84	12.34	12.84
MPT-7B	14.12	14.25	14.33	14.99
Pythia-12B	13.17	12.52	12.08	12.09
Cache	4+508	4+1020	4+2044	4+4092
Llama-2-7B	9.73	9.32	9.08	9.59

5 CONCLUSION

Deploying LLMs in streaming applications is urgently needed but comes with challenges due to efficiency limitations and reduced performance with longer texts. Window attention provides a partial solution, but its performance plummets when initial tokens are excluded. Recognizing the role of these tokens as “attention sinks”, we introduced StreamingLLM—a simple and efficient framework that enables LLMs to handle unlimited texts without fine-tuning. By adding attention sinks with recent tokens, StreamingLLM can efficiently model texts of up to 4 million tokens. We further show that pre-training models with a dedicated sink token can improve the streaming performance. StreamingLLM firstly decouples the LLM’s pre-training window size and its actual text generation length, paving the way for the streaming deployment of LLMs.

REPRODUCIBILITY STATEMENT

All findings presented in this paper can be reproduced. We have made our code and datasets available in this github repo. The models used in this paper are all openly available, and we provide references to access them. Details regarding our experiments, including hyperparameters, training protocols, and evaluation methods, can be found in the Experiments section (Section 4). We are confident that with the provided resources, readers can reproduce the entirety of our presented results.

IMPACT STATEMENT

StreamingLLM has been widely adopted by various LLM serving solutions including NVIDIA TensorRT-LLM, Intel Extension for Transformers, HuggingFace Transformers, MLC LLM, etc.

ACKNOWLEDGEMENTS

This work is supported by MIT-IBM Watson AI Lab, Amazon and MIT Science Hub, National Science Foundation. We thank Angela Li for writing suggestions and demo making, Jingwei Zuo for proofreading, and Xiuyu Li for the suggestion on notations.

REFERENCES

- Joshua Ainslie, Santiago Ontanon, Chris Alberti, Vaclav Cvicek, Zachary Fisher, Philip Pham, Anirudh Ravula, Sumit Sanghai, Qifan Wang, and Li Yang. Etc: Encoding long and structured inputs in transformers, 2020.
- Ebtesam Almazrouei, Hamza Alobeidli, Abdulaziz Alshamsi, Alessandro Cappelli, Ruxandra Cojocaru, Merouane Debbah, Etienne Goffinet, Daniel Heslow, Julien Launay, Quentin Malartic, Badreddine Noune, Baptiste Pannier, and Guilherme Penedo. Falcon-40B: an open large language model with state-of-the-art performance. 2023.
- Sotiris Anagnostidis, Dario Pavllo, Luca Biggio, Lorenzo Noci, Aurelien Lucchi, and Thomas Hofmann. Dynamic context pruning for efficient and interpretable autoregressive transformers, 2023.
- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. Longbench: A bilingual, multitask benchmark for long context understanding. *arXiv preprint arXiv:2308.14508*, 2023.
- Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer, 2020. *arXiv:2004.05150*.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling, 2023.
- Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. Piqa: Reasoning about physical commonsense in natural language. In *Thirty-Fourth AAAI Conference on Artificial Intelligence*, 2020.
- bloc97. NTK-Aware Scaled RoPE allows LLaMA models to have extended (8k+) context size without any fine-tuning and minimal perplexity degradation., 2023. URL https://www.reddit.com/r/LocalLLaMA/comments/14lz7j5/ntkaware_scaled_rope_allows_llama_models_to_have/.
- Yelysei Bondarenko, Markus Nagel, and Tijmen Blankevoort. Quantizable transformers: Removing outliers by helping attention heads do nothing, 2023.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code, 2021.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation, 2023. arXiv: 2306.15595.
- Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality, March 2023. URL <https://lmsys.org/blog/2023-03-30-vicuna/>.
- Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. 2019.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv:1803.05457v1*, 2018.
- Tri Dao. FlashAttention-2: Faster attention with better parallelism and work partitioning. 2023.
- Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness, 2022. arXiv:2205.14135.
- Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers, 2023.
- Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. A dataset of information-seeking questions and answers anchored in research papers, 2021.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *North American Chapter of the Association for Computational Linguistics*, 2019. URL <https://api.semanticscholar.org/CorpusID:52967399>.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021.
- Alexander Fabbri, Irene Li, Tianwei She, Suyi Li, and Dragomir Radev. Multi-news: A large-scale multi-document summarization dataset and abstractive hierarchical model. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019.
- Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. The Pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- Tanya Goyal and Greg Durrett. Evaluating factuality in generation with dependency-level entailment. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, Online, 2020. Association for Computational Linguistics.
- Chi Han, Qifan Wang, Wenhan Xiong, Yu Chen, Heng Ji, and Sinong Wang. LM-Infinite: Simple on-the-fly length generalization for large language models, 2023.

- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, December 2020.
- Luyang Huang, Shuyang Cao, Nikolaus Parulian, Heng Ji, and Lu Wang. Efficient attentions for long document summarization, 2021.
- kaiokendev. Things I’m learning while training superhot., 2023. URL <https://kaiokendev.github.io/til#extending-context-to-8k>.
- Ehsan Kamalloo, Nouha Dziri, Charles L. A. Clarke, and Davood Rafiei. Evaluating open-domain question answering in the era of large language models, 2023.
- Nikita Kitaev, Lukasz Kaiser, and Anselm Levskaya. Reformer: The efficient transformer. In *8th International Conference on Learning Representations, ICLR 2020*. OpenReview.net, April 2020.
- Tomáš Kočiský, Jonathan Schwarz, Phil Blunsom, Chris Dyer, Karl Moritz Hermann, Gábor Melis, and Edward Grefenstette. The narrativeqa reading comprehension challenge, 2017.
- Dacheng Li, Rulin Shao, Anze Xie, Ying Sheng, Lianmin Zheng, Joseph E. Gonzalez, Ion Stoica, Xuezhe Ma, , and Hao Zhang. How long can open-source llms truly promise on context length?, June 2023. URL <https://lmsys.org/blog/2023-06-29-longchat>.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranajpe, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts, 2023.
- Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. Can a suit of armor conduct electricity? a new dataset for open book question answering. In *EMNLP*, 2018.
- Evan Miller. Attention is off by one, 2023. URL <https://www.evanmiller.org/attention-is-off-by-one.html>.
- OpenAI. Gpt-4 technical report, 2023.
- Denis Paperno, Germán Kruszewski, Angeliki Lazaridou, Ngoc Quan Pham, Raffaella Bernardi, Sandro Pezzelle, Marco Baroni, Gemma Boleda, and Raquel Fernández. The LAMBADA dataset: Word prediction requiring a broad discourse context. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1525–1534, Berlin, Germany, August 2016. Association for Computational Linguistics. doi: 10.18653/v1/P16-1144. URL <https://aclanthology.org/P16-1144>.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. Yarn: Efficient context window extension of large language models, 2023.
- Reiner Pope, Sholto Douglas, Aakanksha Chowdhery, Jacob Devlin, James Bradbury, Anselm Levskaya, Jonathan Heek, Kefan Xiao, Shivani Agrawal, and Jeff Dean. Efficiently scaling transformer inference. *arXiv preprint arXiv:2211.05102*, 2022.
- Ofir Press, Noah Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=R8sQPpGCv0>.
- Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever, et al. Improving language understanding by generative pre-training, 2018.
- Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, Chloe Hillier, and Timothy P. Lillicrap. Compressive transformers for long-range sequence modelling. In *International Conference on Learning Representations*, 2020.
- Baptiste Rozière, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, Xiaoqing Ellen Tan, Yossi Adi, Jingyu Liu, Tal Remez, Jérémy Rapin, Artyom Kozhevnikov, Ivan Evtimov, Joanna Bitton, Manish Bhatt, Cristian Canton Ferrer, Aaron Grattafiori, Wenhan Xiong, Alexandre Défossez, Jade Copet, Faisal Azhar, Hugo Touvron, Louis Martin, Nicolas Usunier, Thomas Scialom, and Gabriel Synnaeve. Code Llama: Open foundation models for code, 2023.